

Michael Khoury Computer Engineering Student

 mkhoury@mun.ca  7092198511  St. John's, NL, Canada  LinkedIn  Github  My Portfolio!

EDUCATION

Bachelor of Engineering (Co-op)-Computer Engineering, Class of 2027, Memorial University of Newfoundland St. John's, NL, Canada
Fifth Year, Academic Term 8, GPA: 3.92 / 4.00 (Cumulative), Dean's List

PROFESSIONAL EXPERIENCE

Software Development Student (Co-op), NASDAQ Verafin, Application Team Sep 2026 – Present | St. John's, Canada (Hybrid)
• Developing and designing full-stack, customer-facing features for the Verafin application using object-oriented Java and React
• Leverage AI-assisted development tools including AWS Kiro, GitHub Copilot, and Claude Code to accelerate and optimize development.

Software Development Student (Co-op), NASDAQ Verafin, Check Fraud Infrastructure Team Jan 2026 – Aug 2026 | St. John's, Canada
Co-op (Hybrid): Jan – Apr 2026 | Part-Time (Remote): May – Aug 2026
• Developed and implemented new features in the fraud pipeline using Java, with testing done using JUnit, enhancing pipeline performance
• Built AWS Lambda functions using Python with Pytest testing, and deployed new infrastructure with Terraform in a multi-account AWS environment
• Updated infrastructure configurations and Lambda versions using Terraform and GitHub pull requests, triggering Jenkins CI/CD deployments
• Executed load and QA tests using custom test cases generated through analysis of Splunk trends, uploading load-testing payloads to AWS S3, invoking AWS Step Functions workflows, and analyzing AWS CloudWatch logs/metrics and PostgreSQL SSL database connectivity.
• Migrated Java components' code between repositories, updating dependencies while validating functionality through QA and automated testing.

Systems Engineering Student (Co-op), C-CORE May 2025 – Aug 2025 | St. John's, Canada
• Developed object-oriented C++ and Python signal processing algorithms in Linux for SDR radar systems, improving radar detection accuracy.
• Implemented a real-time filtering C++ block within GNU Radio, ensuring clean, maintainable code structure.
• Built YOLOv8 ML models in Python/PyTorch with custom data preprocessing scripts, delivering 85% accurate automated target detection.

Supplemental Instruction (SI) Leader - ECE (Co-op) Sep 2024 – Dec 2024 | St. John's, Canada
Faculty of Engineering and Applied Science, Memorial University of Newfoundland
• Led weekly SI sessions for 40+ engineering students in C++, Python, digital logic, circuits and semiconductor physics.
• Built a Python-based Excel attendance tracker using Pandas and openpyxl, automating data extraction and saving 5 hours weekly

Machine Learning Software Developer, Visual and Analytic Computing Lab Jun 2024 – Aug 2024 | St. John's, Canada (Hybrid)
• Trained ML models in Python/PyTorch for medical image recognition, running Bash scripts on remote servers via PuTTY, improving accuracy.
• Collaborated in a team of 4 consisting of senior developers and another student to improve ML models, using Git/GitHub, Bitbucket, WinSCP, and a VPN to securely connect to the department's servers, applying version control best practices to streamline development.
• Independently developed a Python script to automate experiment tracking with MLflow, DVC and Bash, improving tracking efficiency.

AI Automation Developer (Co-op), NL Eats Community Outreach Inc Jan 2024 – Apr 2024 | St. John's, Canada (Remote)
• Built AI-driven workflow automations across 5+ projects with the OpenAI API and Zapier. Implemented automated test cases for reliability.

Engineering Co-op Student, Vale Canada Ltd May 2023 – Aug 2023 | Long Harbour, Canada
• Analyzed equipment data and maintenance schedules with Excel/SAP to support preventative maintenance plans, improving uptime by 15%.

PROJECTS

Bank Transaction Monitoring System, Personal Project Jul 2025 – Present
• Built a Bank Transaction Monitoring System with Java, Spring Boot, PostgreSQL, and Maven, creating RESTful APIs with clean code.
• Implemented real-time fraud detection with Apache Spark, using distributed computing on AWS EC2/RDS, and tested APIs using Postman.

Full-Stack Portfolio Website, Personal Project May 2025 – Jun 2025
• Built a portfolio site in JavaScript (React.js) and CSS with a Node.js/Express.js and MongoDB REST API backend that includes a contact form.

Tower Defense Game (Unity and C#), Software Design Course Jan 2025 – Apr 2025

SKILLS

Languages: Python, C++, C, C#, VHDL, JavaScript, Java, Terraform, HTML, CSS, SQL, YAML, UML, JSON, XML, Bash, Dafny, Assembly, MATLAB
Frameworks: JUnit, React.js, Node.js, Express.js, Spring Boot, .NET, Unity, PyTorch, Pytest, NumPy, Pandas, Matplotlib, openpyxl, OpenCV
Tools: AWS, Git/GitHub/GitLab, Jenkins, Gradle, Splunk, Salesforce, Jira/Bitbucket, Notion, Confluence, VS Code, Google CoLab, MongoDB, Conda, Zapier, Shell/SSH, WinSCP, PuTTY, Postman, IntelliJ IDEA, Apache Spark, PostgreSQL, DynamoDB, Cmake, Simulink, DVC, MLflow

AWARDS

Faculty of Engineering and Applied Science Dean's List 24/25, 25/26, Memorial University of Newfoundland	2025
International Academic Award (\$3400), Memorial University of Newfoundland	2025
IUGS Entrance Scholarship (\$4000), Memorial University of Newfoundland	2022
Class of 2021 High School Valedictorian, The English Modern School, Doha	2021